

MarketGyan: NEPSE-Impact Dataset, Model Results, and RAG Demo Progress

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Project Supervisor Discussion

June 21, 2026

Project Goal

Research contribution

MarketGyan builds a Nepal-specific benchmark and modeling workflow for deciding whether news is relevant to NEPSE, mapping it to sectors and listed symbols, and producing evidence-grounded market-impact labels.

- Core dataset: human-adjudicated bilingual **NEPSE-Impact-500**.
- Modeling: XLM-R baseline, FinBERT English subset baseline, and Qwen3-8B QLoRA experiment.
- Runtime principle: current market facts come from deterministic data and RAG, not from model memory.
- Output boundary: informational analysis only, never investment advice or trading signals.

Why the Problem Was Reframed

What failed in the first idea

General business news looked financial but often had no defensible NEPSE impact mechanism.

- Non-listed company announcements
- Product promotions and workshops
- Foreign business or technology news
- Broad economy stories without market linkage

Revised task

Teach the model Nepal-market relevance, event taxonomy, entity mapping, and evidence selection.

- Direct listed-company or market impact
- Indirect sector or macro mechanism
- Hard-negative general business examples

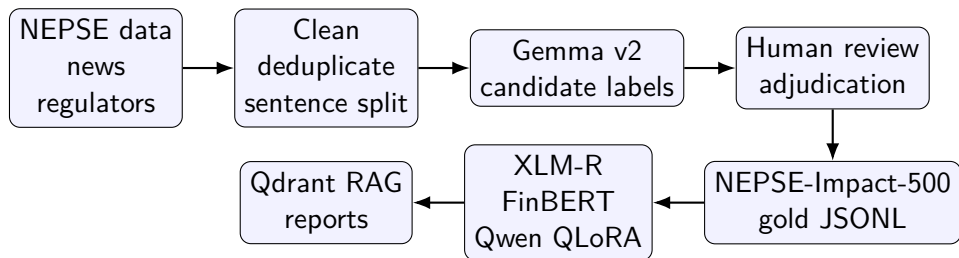
Novelty Position

Reference idea	MarketGyan contribution
FinBERT	Domain adaptation, but focused on Nepal market relevance instead of generic English finance sentiment.
FinGPT	Data-centric lightweight tuning using a local, auditable, human-reviewed finance corpus.
FinLLM survey	Benchmark design with hallucination, validity, grounding, language, and event-category metrics.
FinTral	Multi-task financial instruction tuning with structured labels, sectors, symbols, and evidence.

Supervisor-facing claim

LoRA is not the novelty. The novelty is the NEPSE-specific ontology, bilingual benchmark, adjudication workflow, and evidence-grounded evaluation.

End-to-End Data Pipeline



- Numeric market snapshots are deterministic and stored with provenance.
- Gemma is only a teacher candidate generator; human adjudication creates gold labels.

Corpus Status: NEPSE-Impact-500

Component	Count	Status	Purpose
Gold records	500	Ready	Training and evaluation dataset.
Direct records	300	Target met	Listed company or market impact.
Indirect records	100	Target met	Sector or macro mechanism.
Hard negatives	100	Target met	Business news without NEPSE mechanism.
English records	299	Target met	English benchmark coverage.
Nepali records	200	Target met	Nepali benchmark coverage.
Symbol-level records	273	Target met	Listed-company mapping task.

Quality control

Duplicate groups are isolated in splits; records retain source URL, date, candidate label, human label, and evidence sentence IDs.

Ontology and Gold Label

Classification fields

- Relevance: direct, indirect, not relevant
- Event type
- Impact scope
- Impact direction
- Impact horizon
- Impact mechanism

Grounding fields

- NEPSE sector names
- Listed symbols
- Confidence band
- Evidence sentence IDs
- Source metadata and URL

Annotation rule

A positive label requires a concrete NEPSE company, sector, market, regulatory, liquidity, or macroeconomic mechanism.

Current Baseline Results

Model/task	Accuracy	Macro-F1	Interpretation
XLM-R relevance	0.787	0.723	Current dependable relevance baseline.
XLM-R direction	0.683	0.549	Usable, but direction classes still need more data.
FinBERT English direction	0.889	0.616	Strong on small English relevant subset; not bilingual.
Qwen3 QLoRA v3 strict	0.333	0.384	Failed official strict-output gate.

Current conclusion

XLM-R is the strongest reliable classifier. Qwen is still a research experiment until strict JSON validity and evidence grounding improve.

Qwen3-8B QLoRA: What We Learned

Metric	Strict v3	Tolerant diagnostic	Meaning
JSON validity	36.0%	68.0%	Many outputs are near-valid, not production-valid.
Evidence grounding	36.0%	68.0%	Sentence IDs improve only after repair.
Relevance macro-F1	0.384	0.555	The adapter learned some relevance signal.
Direction macro-F1	0.413	0.629	Direction extraction is promising but not deployable.
Symbol F1	–	0.790	Entity extraction is the strongest Qwen signal.
Evidence F1	–	0.633	Evidence selection still needs control.

Important boundary

Tolerant repair is failure analysis only. It does not count as official validity or deployment readiness.

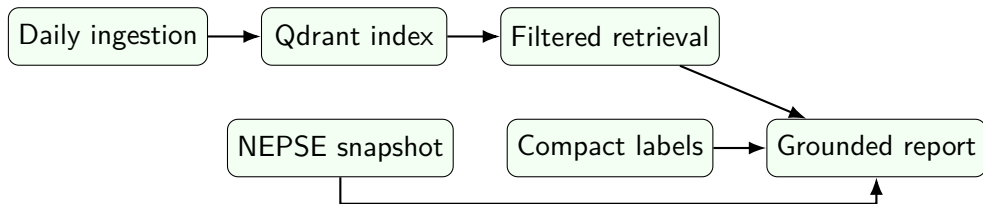
Why Qwen Is Not Ready Yet

- Fine-tuning raises the probability of schema-like text but cannot guarantee JSON.
- The model often emits unquoted keys, Markdown/prose, repeated IDs, or non-canonical enums.
- Nepali and hard-negative examples remain more difficult than direct English finance news.
- Runtime use needs schema-constrained decoding through vLLM or an OpenAI-compatible JSON Schema path.

Decision

Keep Qwen as a compact classifier/extractor. Do not use it as a free-form post-market report writer.

RAG and Report Generation Design



Citation rule

Sentence IDs are internal anchors only. User-facing reports must show source title, URL, date, chunk ID, content hash, and exact evidence text.

Demo-safe mode

A deterministic FastAPI mock agent can build answers and reports from retrieved Qdrant citations while Qwen remains disabled for runtime claims.

Implemented Demo Workspace

UI area	Current behavior
Overview	Snapshot mini-cards, sector/story placeholders, latest report summary, disclaimer.
Ask MarketGyan	Grounded Q&A with filters, citations, disclaimer, and fail-closed errors.
Evidence Search	Qdrant search with source URLs, scores, sectors, symbols, and sentence anchors.
Reports	Latest published report, citation cards, and local-only generation control.
System	Runtime readiness checklist with safe booleans only; no secrets exposed.
Validate data	Local-review tab remains available only when review API is enabled.

Recent UI hardening

Forms are disabled when runtime status is unavailable, report generation blocks future dates, and snapshot metrics no longer collide visually.

Current Demonstration Path

- 1 Show NEPSE-Impact-500 and the schema-v2 ontology.
- 2 Show XLM-R, FinBERT, and Qwen v3 results with the strict-output limitation.
- 3 Start Qdrant and index MarketGyan evidence.
- 4 Start Node backend with query and review flags enabled.
- 5 Start FastAPI agent in deterministic mock mode.
- 6 Use the dashboard: Evidence Search, Ask MarketGyan, Reports, and System.
- 7 Verify every answer/report includes citations and the disclaimer.

Remaining Work

- 1 Complete second-review agreement on 110 stratified records.
- 2 Evaluate at least 30 retrieval queries and 20 report/query scenarios.
- 3 Add schema-constrained Qwen decoding for runtime-style evaluation.
- 4 Decide whether Qwen is a positive result or negative experiment.
- 5 Improve direction-class coverage if final model metrics are weak.
- 6 Run the final supervisor demo with Qdrant, Node, FastAPI, and React.

Deployment gate

Runtime Qwen use remains disabled unless strict JSON validity and evidence grounding reach at least 95%.

Supervisor Discussion Points

What is complete

Data pipeline, ontology, 500 adjudicated records, baseline experiments, Qwen v3 failure analysis, RAG citation design, FastAPI mock agent, and full dashboard demo workspace.

What needs feedback

Whether to present Qwen as a negative experiment unless constrained decoding passes, and how much second-review agreement is required before final submission.

Core message

MarketGyan is an evidence-grounded Nepal market-impact benchmark and demo system, not a prediction engine or investment-advice tool.